

Corrected Bile Acid Analysis Report

Fixed and verified CSV/R-derived longitudinal mixed-model analysis of fasting and postprandial bile-acid dynamics. Source data: IRMA23930_MS-BA_Results.csv. Generated from corrected scripts in comparison_work.

Statistical Methods Guide

This report uses a longitudinal linear mixed-effects model because each subject contributes repeated bile-acid measurements across multiple timepoints. A standard linear model or independent t-test would ignore within-subject correlation. The model used for each bile acid or aggregate class was **log(concentration) ~ timepoint + (1 | subject)**, where timepoint is the fixed effect of interest and subject is a random intercept that accounts for each participant's baseline level.

Term	Meaning in this report
Log transform	Concentrations were analyzed on the natural-log scale. This stabilizes variance and makes skewed concentration data more suitable for mixed-model inference.
LFC	Log fold change. Here it is the model-estimated difference on the natural-log scale, such as $\log(23) - \log(2)$. Positive LFC means the post-meal timepoint is higher; negative LFC means it is lower.
Fold	Back-transformed LFC: $\text{Fold} = \exp(\text{LFC})$. A fold of 1.50 means 50% higher; 0.75 means 25% lower.
EMM	Estimated marginal mean from the LMM. These are model-adjusted means for each timepoint, back-transformed to the concentration scale for plots.
Satterthwaite	A finite-sample degrees-of-freedom approximation for mixed models. It gives more appropriate p-values for moderate repeated-measures datasets than treating test statistics as if sample size were infinite.
BH FDR	Benjamini-Hochberg false discovery rate correction. Because many contrasts are tested, raw p-values are adjusted to q-values to control the expected proportion of false discoveries among significant results.
q-value	The FDR-adjusted p-value. This report calls results significant using $q < 0.05$, not raw $p < 0.05$.

Contrasts And Significance Rules

Analysis	How it was done
Main contrasts	Timepoint 23 was compared separately against baseline fasting (2), NS4 fasting (18), and RTDS fasting (26). These are shown as 23 vs 2, 23 vs 18, and 23 vs 26.
Multiple testing families	FDR correction was applied separately to the 45 individual bile-acid contrasts (15 bile acids x 3 contrasts) and the 12 aggregate-class contrasts (4 aggregate classes x 3 contrasts).
Significance stars	Stars are based on FDR q-values: *** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$, ns = not significant.
Plot layout	Plots use the requested divergence layout: blue branch 2->18->26 and orange branch 18->23->26. Only the timepoint 23 marker is orange because it is the meal branch point.
Missing values	Missing values were not imputed. The all-missing sample 1623-26 was kept missing for aggregate classes rather than converted to zero.

What Was Fixed

Problem in prior PDF	Correction in this report
Raw p-value stars were used for interpretation.	Tables and text use BH FDR q-values for significance.
Aggregate rowSums(..., na.rm=TRUE) converted an all-missing aggregate row to NA.	All missing aggregate rows are preserved as NA and excluded compound-by-compound.
Model df method was not explicit.	emmeans is called with lmer.df='satterthwaite' for finite-sample inference.
Narrative overclaimed significance for several aggregate classes.	Interpretation text is generated from the corrected q-values.

Corrected Statistical Method

For each individual bile acid and each aggregate class, adjusted positive concentrations were modeled as **log(concentration) ~ timepoint + (1 | subject)**. Timepoint 23 was compared with baseline fasting (2), NS4 fasting (18), and RTDS fasting (26). P-values use Satterthwaite finite-sample degrees of freedom. BH FDR correction was applied separately to the 45 individual-bile-acid contrasts and the 12 aggregate-class contrasts.

Verification Checks

Check	Result
Raw data rows	203
Biological sample rows	159
Longitudinal rows used before compound-specific missing filtering	127
Subjects	32
Timepoint 10 excluded	Yes
All-missing BA rows preserved as missing	1 row: 1623-26
Finite-sample df method	Satterthwaite via emmeans(lmer.df = 'satterthwaite')
Multiple testing correction	BH FDR, separately for 45 individual BA contrasts and 12 aggregate contrasts
Non-positive modeled values	0

High-Level Corrected Result

The strongest corrected postprandial signals remain concentrated in conjugated bile acids. Against baseline fasting (2), GCDCA, TCDCA, GLCA, TLCA, GCA, TCA, and TDCA are FDR-significant. Against NS4 fasting (18), TCDCA, LCA, GLCA, TLCA, TCA, and TDCA are FDR-significant. Against RTDS fasting (26), only LCA and GLCA survive FDR among individual bile acids.

Master Results Table: 23 vs Baseline Fasting (2)

Compound	Class	Family	N	LFC	Fold	raw p	FDR q	Sig
GCA	Primary Conjugated	Individual BA	126	0.505	1.66	<0.001	0.002	**
TCA	Primary Conjugated	Individual BA	126	0.654	1.92	<0.001	<0.001	***
GCDCA	Primary Conjugated	Individual BA	126	0.494	1.64	<0.001	0.002	**
TCDCA	Primary Conjugated	Individual BA	126	0.814	2.26	<0.001	<0.001	***
CA	Primary Unconjugated	Individual BA	126	0.429	1.54	0.101	0.228	ns
CDCA	Primary Unconjugated	Individual BA	126	0.071	1.07	0.794	0.871	ns
GDCA	Secondary Conjugated	Individual BA	121	0.154	1.17	0.321	0.481	ns
TDCA	Secondary Conjugated	Individual BA	120	0.600	1.82	<0.001	0.002	**
GLCA	Secondary Conjugated	Individual BA	124	0.611	1.84	<0.001	<0.001	***
TLCA	Secondary Conjugated	Individual BA	121	0.465	1.59	0.012	0.040	*
DCA	Secondary Unconjugated	Individual BA	121	-0.334	0.72	0.053	0.141	ns
LCA	Secondary Unconjugated	Individual BA	120	-0.100	0.91	0.281	0.469	ns
UDCA	Tertiary / Microbial	Individual BA	103	-0.184	0.83	0.411	0.561	ns
GUDCA	Tertiary / Microbial	Individual BA	126	-0.077	0.93	0.516	0.645	ns
TUDCA	Tertiary / Microbial	Individual BA	125	0.194	1.21	0.171	0.321	ns
Total_BA	Aggregate Total Pool	Aggregate class	126	0.424	1.53	0.002	0.007	**
Glycine_Conjugated	Aggregate Glycine Conjugates	Aggregate class	126	0.415	1.51	0.002	0.007	**
Taurine_Conjugated	Aggregate Taurine Conjugates	Aggregate class	126	0.713	2.04	<0.001	<0.001	***
Unconjugated	Aggregate Unconjugated Pool	Aggregate class	126	0.045	1.05	0.842	0.918	ns

Master Results Table: 23 vs NS4 Fasting (18)

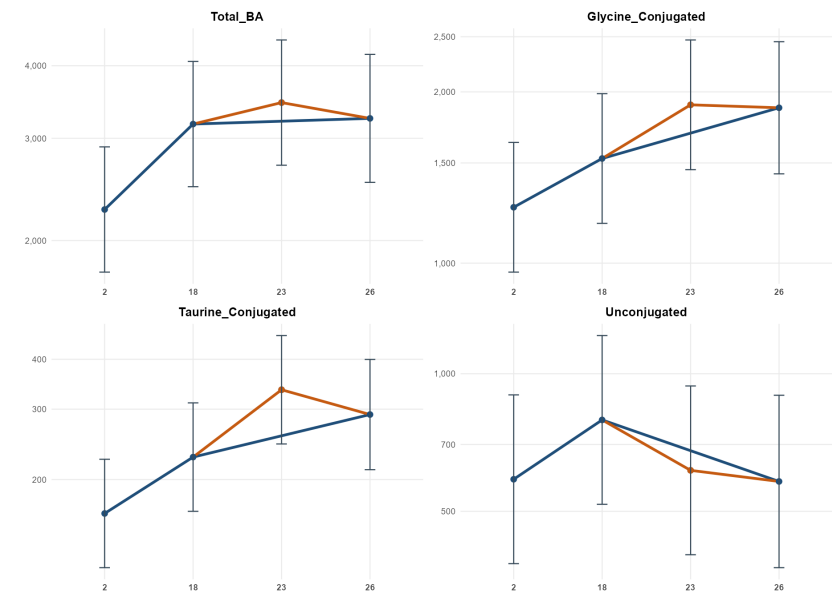
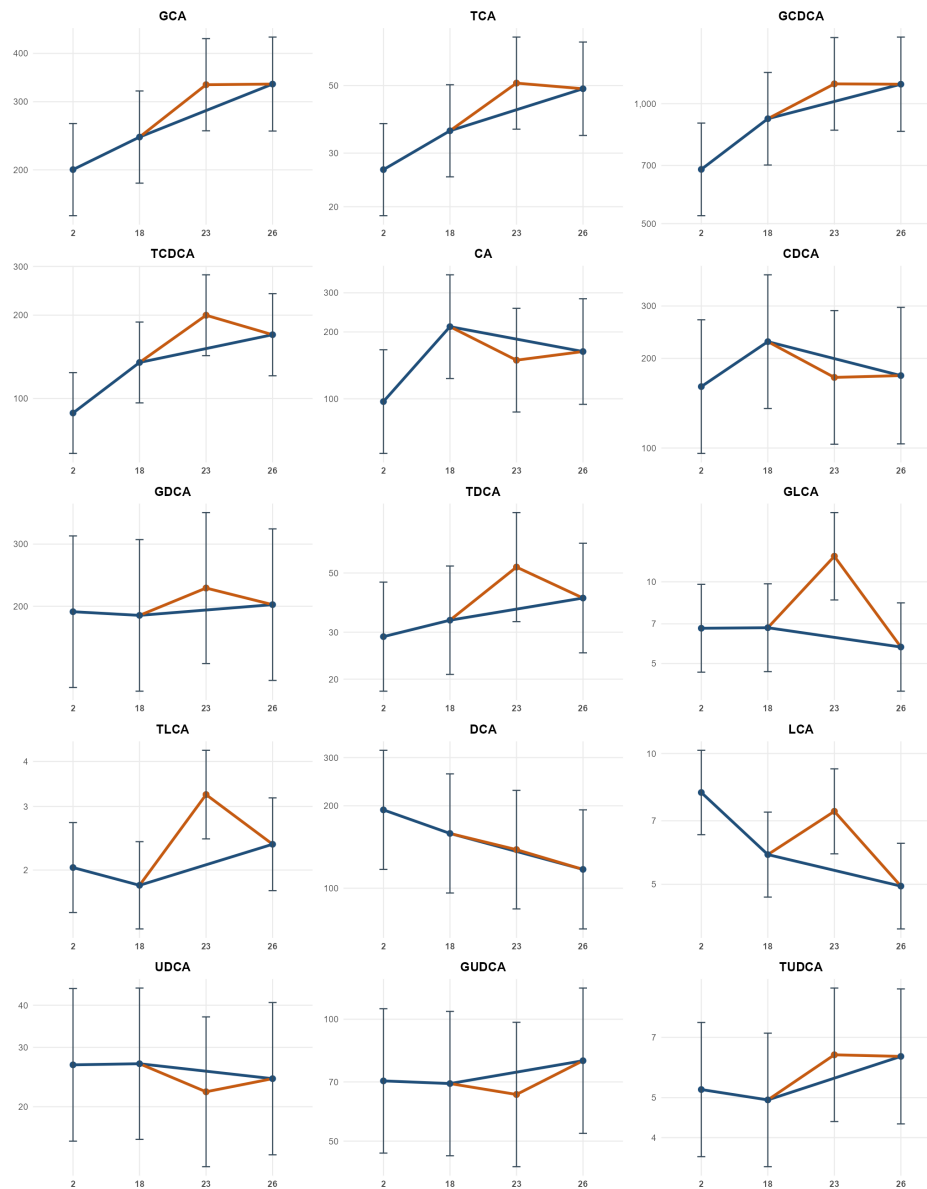
Compound	Class	Family	N	LFC	Fold	raw p	FDR q	Sig
GCA	Primary Conjugated	Individual BA	126	0.312	1.37	0.025	0.071	ns
TCA	Primary Conjugated	Individual BA	126	0.360	1.43	0.016	0.048	*
GCDCA	Primary Conjugated	Individual BA	126	0.201	1.22	0.123	0.240	ns
TCDCa	Primary Conjugated	Individual BA	126	0.393	1.48	0.011	0.040	*
CA	Primary Unconjugated	Individual BA	126	-0.347	0.71	0.184	0.331	ns
CDCA	Primary Unconjugated	Individual BA	126	-0.277	0.76	0.311	0.481	ns
GDCA	Secondary Conjugated	Individual BA	121	0.178	1.20	0.252	0.436	ns
TDCA	Secondary Conjugated	Individual BA	120	0.458	1.58	0.006	0.026	*
GLCA	Secondary Conjugated	Individual BA	124	0.606	1.83	<0.001	<0.001	***
TLCA	Secondary Conjugated	Individual BA	121	0.579	1.78	0.002	0.007	**
DCA	Secondary Unconjugated	Individual BA	121	-0.136	0.87	0.427	0.565	ns
LCA	Secondary Unconjugated	Individual BA	120	0.229	1.26	0.016	0.048	*
UDCA	Tertiary / Microbial	Individual BA	103	-0.192	0.83	0.382	0.538	ns
GUDCA	Tertiary / Microbial	Individual BA	126	-0.062	0.94	0.602	0.732	ns
TUDCA	Tertiary / Microbial	Individual BA	125	0.252	1.29	0.074	0.184	ns
Total_BA	Aggregate Total Pool	Aggregate class	126	0.085	1.09	0.518	0.777	ns
Glycine_Conjugated	Aggregate Glycine Conjugates	Aggregate class	126	0.217	1.24	0.093	0.224	ns
Taurine_Conjugated	Aggregate Taurine Conjugates	Aggregate class	126	0.388	1.47	0.009	0.026	*
Unconjugated	Aggregate Unconjugated Pool	Aggregate class	126	-0.254	0.78	0.261	0.521	ns

Master Results Table: 23 vs RTDS Fasting (26)

Compound	Class	Family	N	LFC	Fold	raw p	FDR q	Sig
GCA	Primary Conjugated	Individual BA	126	-0.004	1.00	0.978	0.988	ns
TCA	Primary Conjugated	Individual BA	126	0.042	1.04	0.777	0.871	ns
GCDCA	Primary Conjugated	Individual BA	126	0.002	1.00	0.988	0.988	ns
TCDCA	Primary Conjugated	Individual BA	126	0.162	1.18	0.297	0.477	ns
CA	Primary Unconjugated	Individual BA	126	-0.089	0.91	0.736	0.850	ns
CDCA	Primary Unconjugated	Individual BA	126	-0.014	0.99	0.960	0.988	ns
GDCA	Secondary Conjugated	Individual BA	121	0.108	1.11	0.485	0.624	ns
TDCA	Secondary Conjugated	Individual BA	120	0.267	1.31	0.111	0.238	ns
GLCA	Secondary Conjugated	Individual BA	124	0.770	2.16	<0.001	<0.001	***
TLCA	Secondary Conjugated	Individual BA	121	0.317	1.37	0.089	0.211	ns
DCA	Secondary Unconjugated	Individual BA	121	0.166	1.18	0.333	0.484	ns
LCA	Secondary Unconjugated	Individual BA	120	0.396	1.49	<0.001	<0.001	***
UDCA	Tertiary / Microbial	Individual BA	103	-0.090	0.91	0.691	0.819	ns
GUDCA	Tertiary / Microbial	Individual BA	126	-0.192	0.83	0.116	0.238	ns
TUDCA	Tertiary / Microbial	Individual BA	125	0.009	1.01	0.950	0.988	ns
Total_BA	Aggregate Total Pool	Aggregate class	126	0.063	1.06	0.641	0.854	ns
Glycine_Conjugated	Aggregate Glycine Conjugates	Aggregate class	126	0.012	1.01	0.927	0.927	ns
Taurine_Conjugated	Aggregate Taurine Conjugates	Aggregate class	126	0.143	1.15	0.336	0.576	ns
Unconjugated	Aggregate Unconjugated Pool	Aggregate class	126	0.056	1.06	0.808	0.918	ns

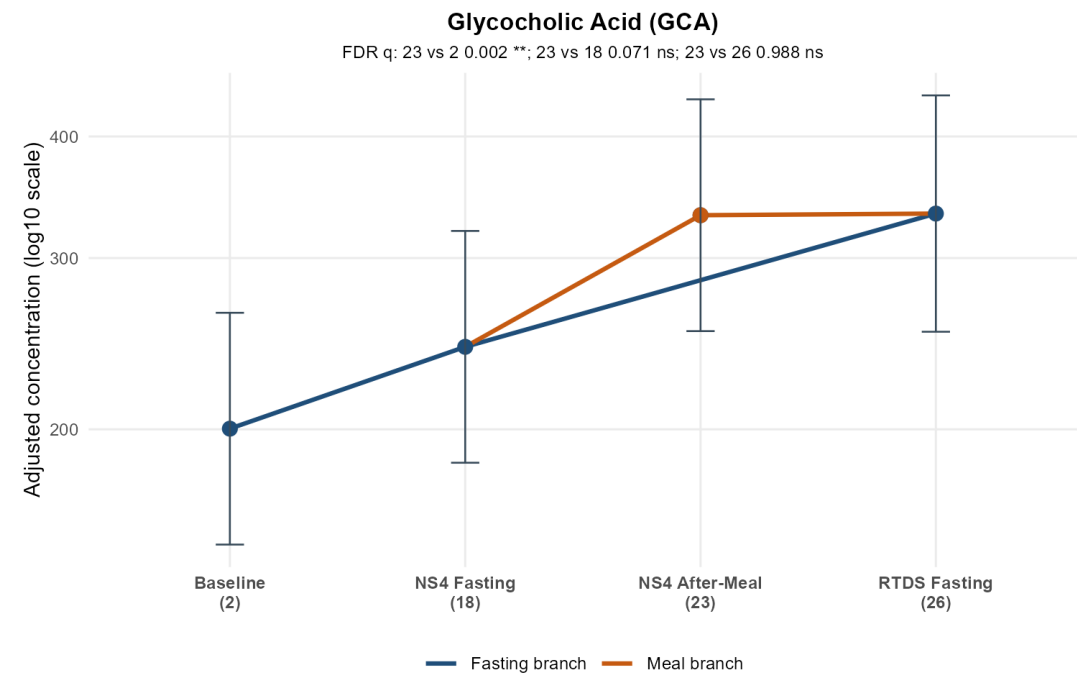
Trajectory Overview

Plots show LMM estimated marginal means back-transformed to the concentration scale. Error bars are 95% confidence intervals. All trajectories use a divergence layout: the fasting branch runs from baseline (2) through NS4 fasting (18) to RTDS fasting (26), while the meal branch starts at NS4 fasting (18), passes through NS4 after-meal (23), and merges again at RTDS fasting (26). The y-axis is log10-scaled for readability; significance labels in the detailed plots use FDR q-values.



GCA: Glycocholic Acid (GCA)

Primary Conjugated | Individual BA | N=126, subjects=32



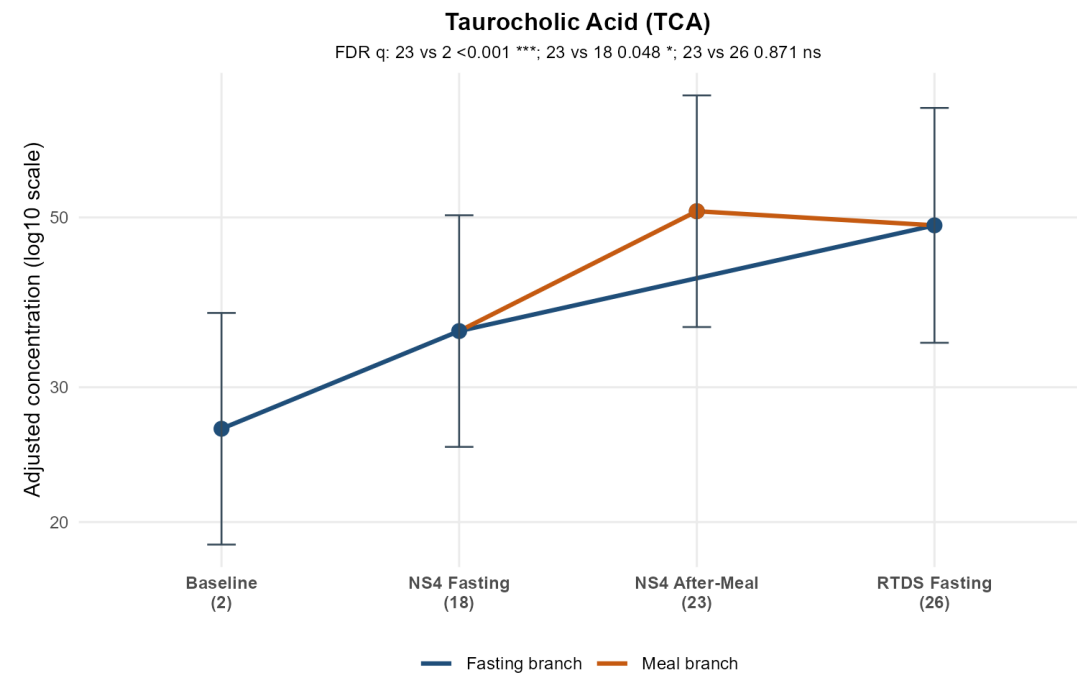
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.505	1.66	<0.001	0.002	**
23 vs 18	0.312	1.37	0.025	0.071	ns
23 vs 26	-0.004	1.00	0.978	0.988	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=0.002, **).

TCA: Taurocholic Acid (TCA)

Primary Conjugated | Individual BA | N=126, subjects=32



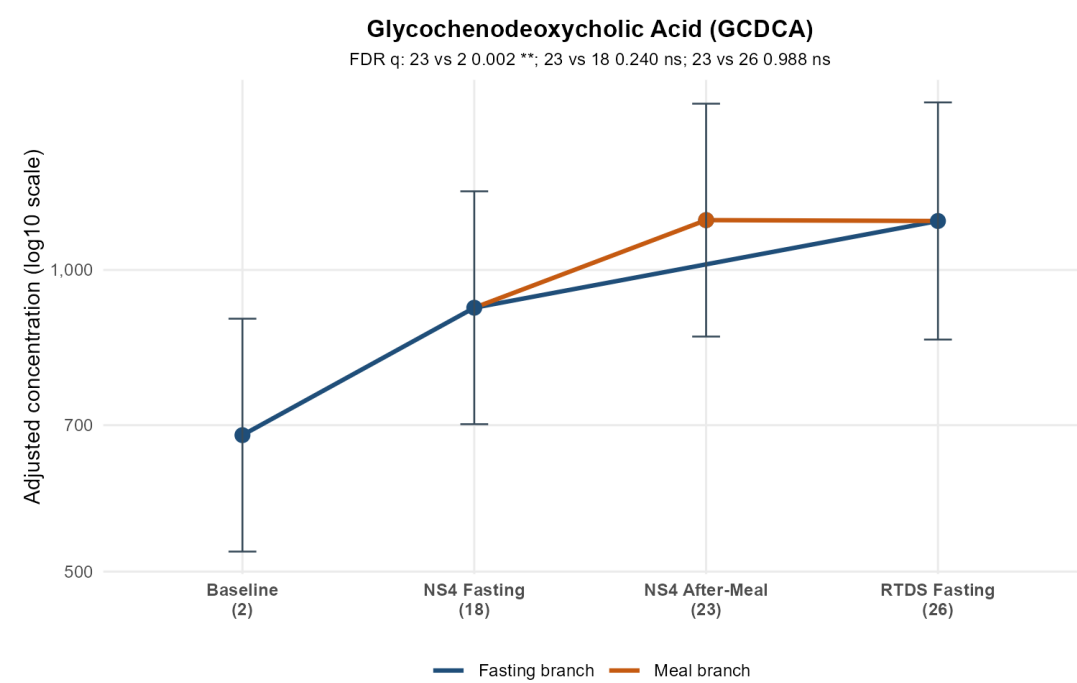
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.654	1.92	<0.001	<0.001	***
23 vs 18	0.360	1.43	0.016	0.048	*
23 vs 26	0.042	1.04	0.777	0.871	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=<0.001, ***); higher than NS4 fasting (18) (FDR q=0.048, *).

GCDCA: Glycochenodeoxycholic Acid (GCDCA)

Primary Conjugated | Individual BA | N=126, subjects=32



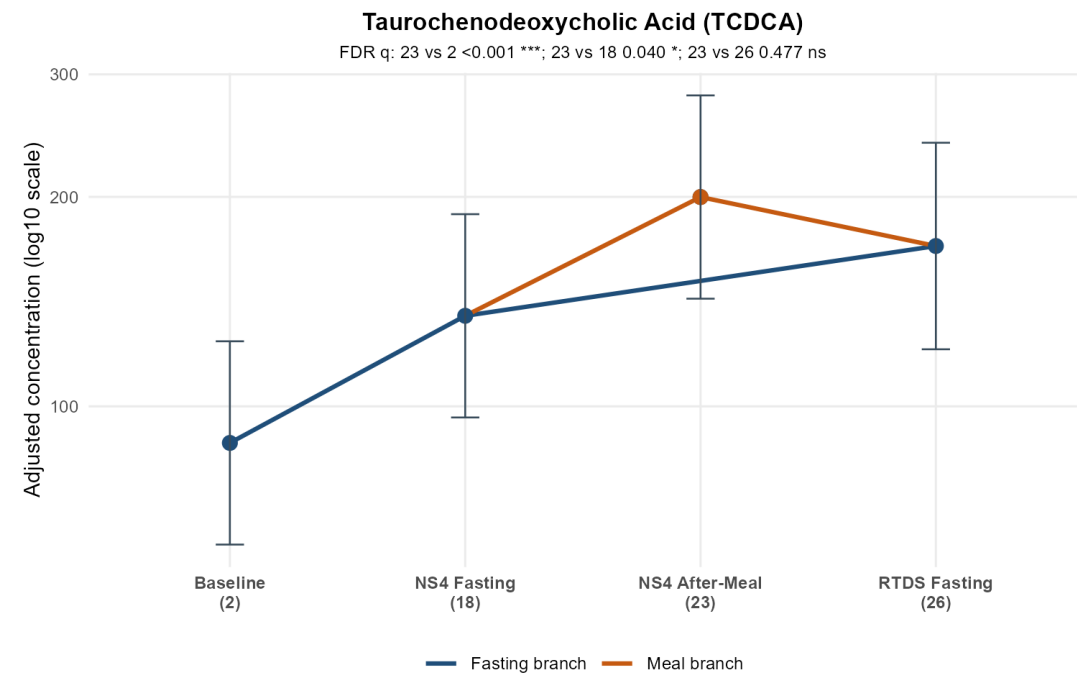
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.494	1.64	<0.001	0.002	**
23 vs 18	0.201	1.22	0.123	0.240	ns
23 vs 26	0.002	1.00	0.988	0.988	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=0.002, **).

TCDCA: Taurochenodeoxycholic Acid (TCDCA)

Primary Conjugated | Individual BA | N=126, subjects=32

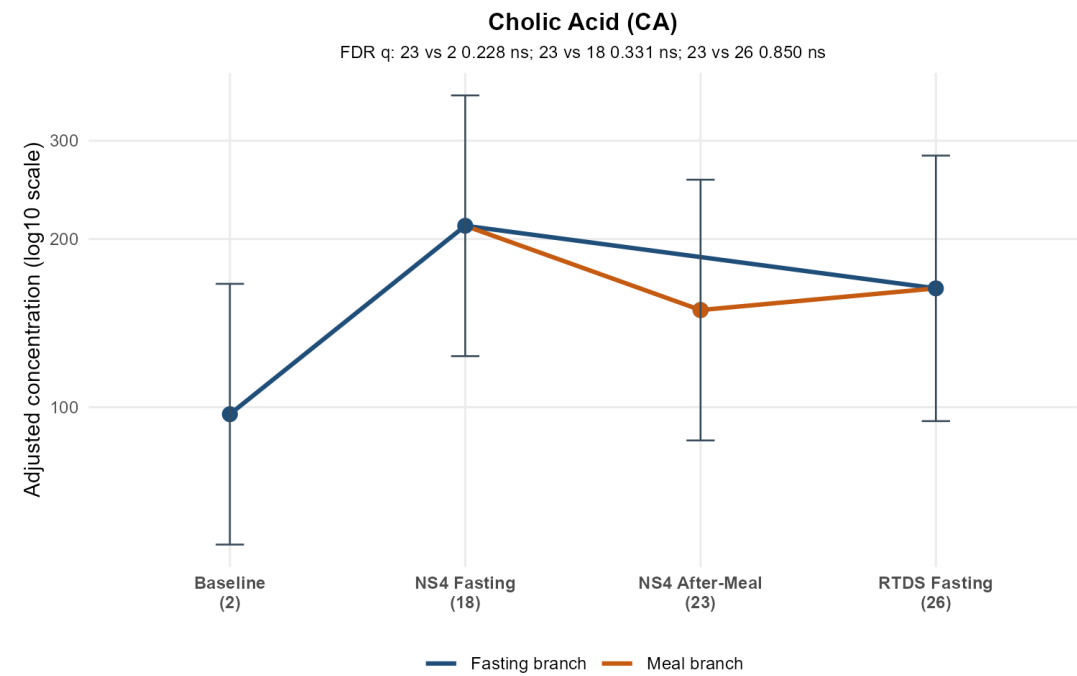


Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.814	2.26	<0.001	<0.001	***
23 vs 18	0.393	1.48	0.011	0.040	*
23 vs 26	0.162	1.18	0.297	0.477	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=<0.001, ***); higher than NS4 fasting (18) (FDR q=0.040, *).

CA: Cholic Acid (CA)

Primary Unconjugated | Individual BA | N=126, subjects=32

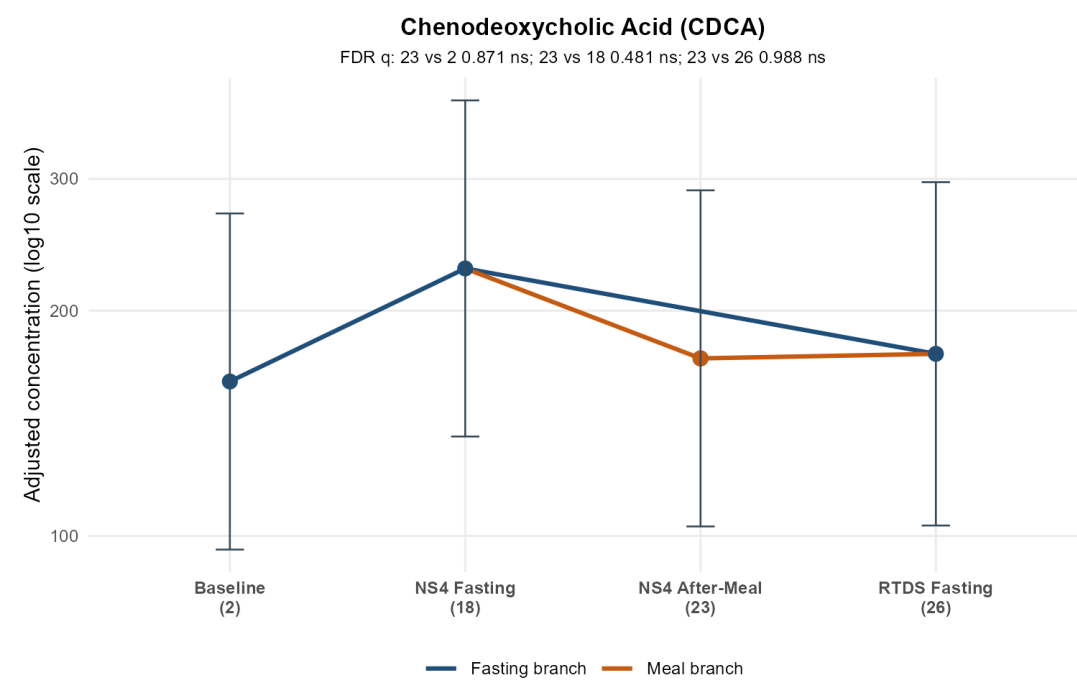


Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.429	1.54	0.101	0.228	ns
23 vs 18	-0.347	0.71	0.184	0.331	ns
23 vs 26	-0.089	0.91	0.736	0.850	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

CDCA: Chenodeoxycholic Acid (CDCA)

Primary Unconjugated | Individual BA | N=126, subjects=32



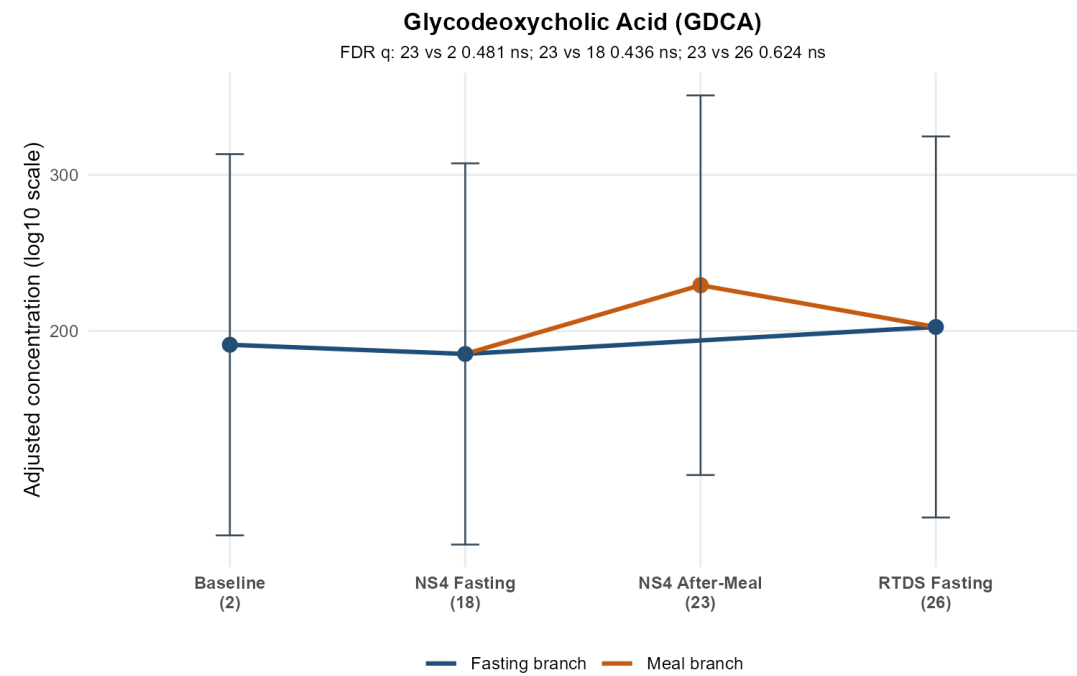
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.071	1.07	0.794	0.871	ns
23 vs 18	-0.277	0.76	0.311	0.481	ns
23 vs 26	-0.014	0.99	0.960	0.988	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

GDCA: Glycodeoxycholic Acid (GDCA)

Secondary Conjugated | Individual BA | N=121, subjects=31



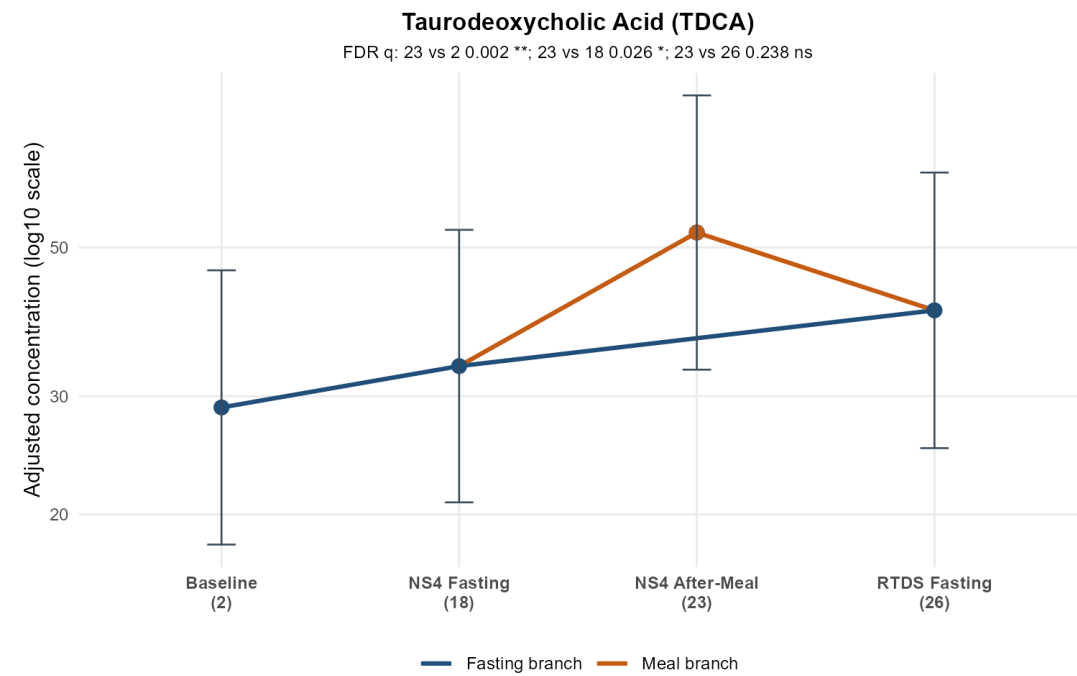
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.154	1.17	0.321	0.481	ns
23 vs 18	0.178	1.20	0.252	0.436	ns
23 vs 26	0.108	1.11	0.485	0.624	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

TDCA: Taurodeoxycholic Acid (TDCA)

Secondary Conjugated | Individual BA | N=120, subjects=31



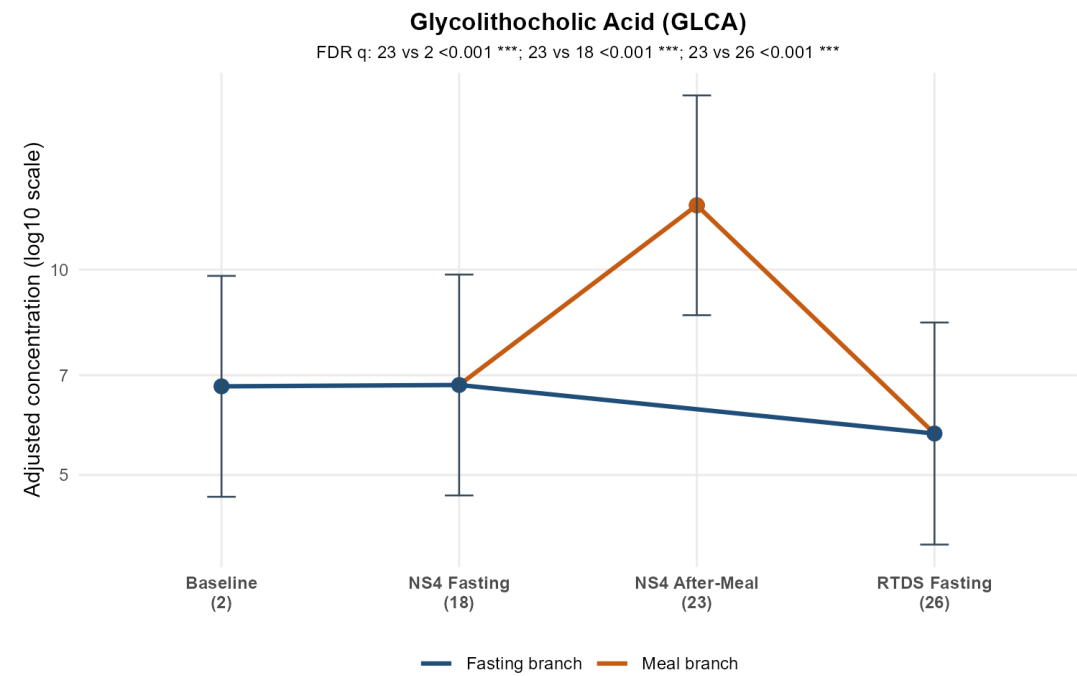
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.600	1.82	<0.001	0.002	**
23 vs 18	0.458	1.58	0.006	0.026	*
23 vs 26	0.267	1.31	0.111	0.238	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=0.002, **); higher than NS4 fasting (18) (FDR q=0.026, *).

GLCA: Glycolithocholic Acid (GLCA)

Secondary Conjugated | Individual BA | N=124, subjects=32

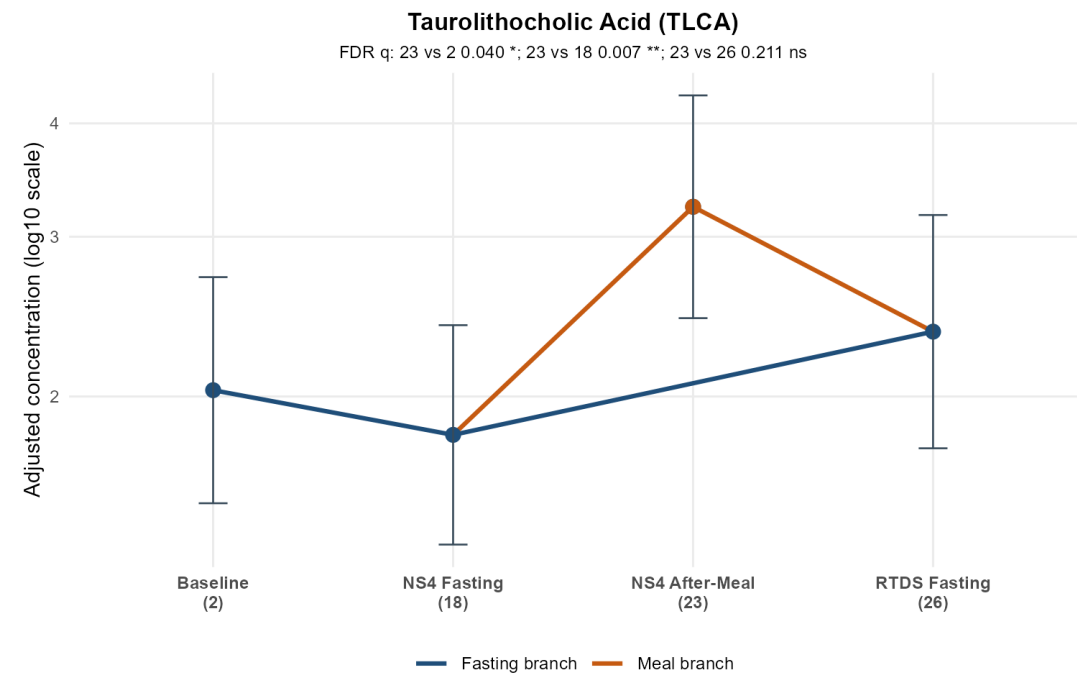


Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.611	1.84	<0.001	<0.001	***
23 vs 18	0.606	1.83	<0.001	<0.001	***
23 vs 26	0.770	2.16	<0.001	<0.001	***

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=<0.001, ***); higher than NS4 fasting (18) (FDR q=<0.001, ***); higher than RTDS fasting (26) (FDR q=<0.001, ***).

TLCA: Tauroolithocholic Acid (TLCA)

Secondary Conjugated | Individual BA | N=121, subjects=32

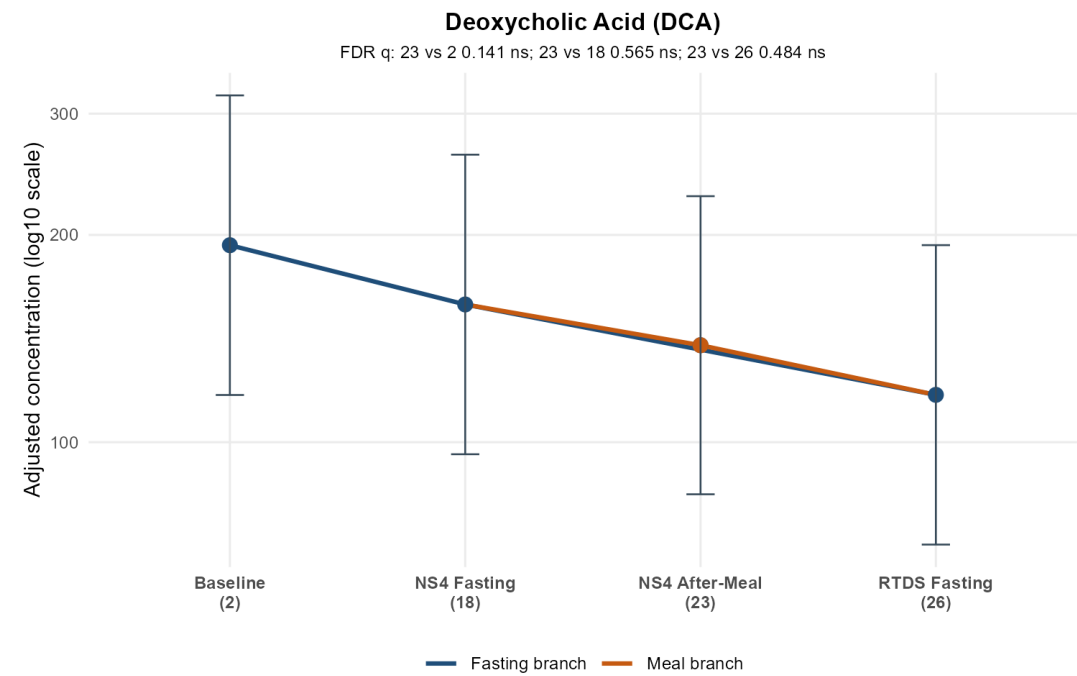


Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.465	1.59	0.012	0.040	*
23 vs 18	0.579	1.78	0.002	0.007	**
23 vs 26	0.317	1.37	0.089	0.211	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=0.040, *); higher than NS4 fasting (18) (FDR q=0.007, **).

DCA: Deoxycholic Acid (DCA)

Secondary Unconjugated | Individual BA | N=121, subjects=31



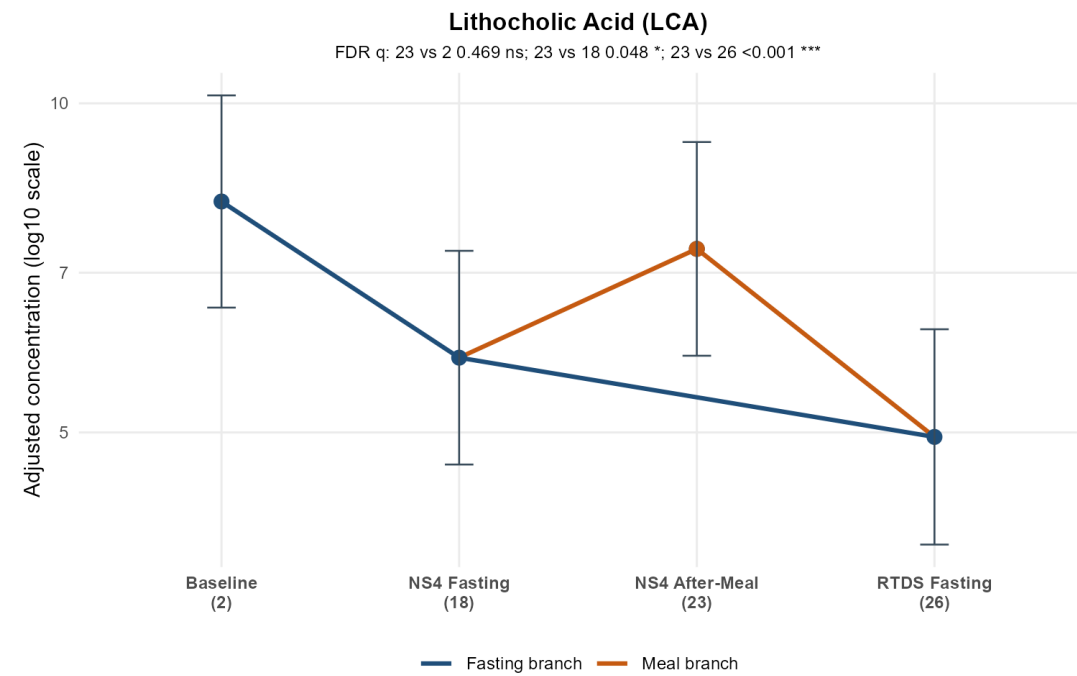
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	-0.334	0.72	0.053	0.141	ns
23 vs 18	-0.136	0.87	0.427	0.565	ns
23 vs 26	0.166	1.18	0.333	0.484	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

LCA: Lithocholic Acid (LCA)

Secondary Unconjugated | Individual BA | N=120, subjects=31

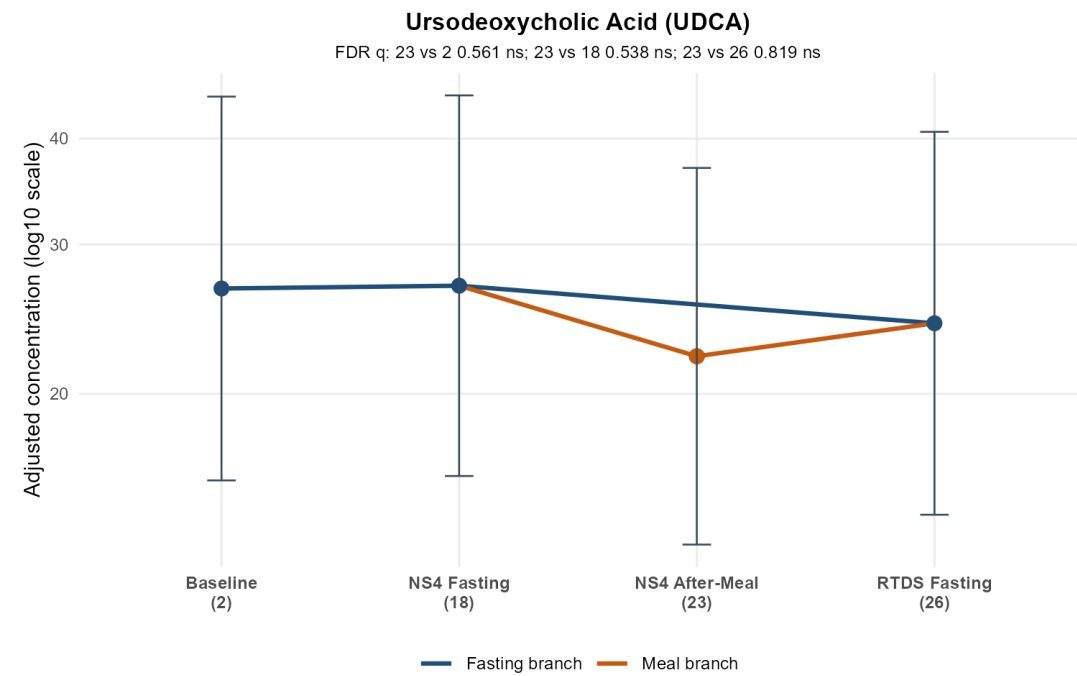


Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	-0.100	0.91	0.281	0.469	ns
23 vs 18	0.229	1.26	0.016	0.048	*
23 vs 26	0.396	1.49	<0.001	<0.001	***

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than NS4 fasting (18) (FDR q=0.048, *); higher than RTDS fasting (26) (FDR q=<0.001, ***).

UDCA: Ursodeoxycholic Acid (UDCA)

Tertiary / Microbial | Individual BA | N=103, subjects=30



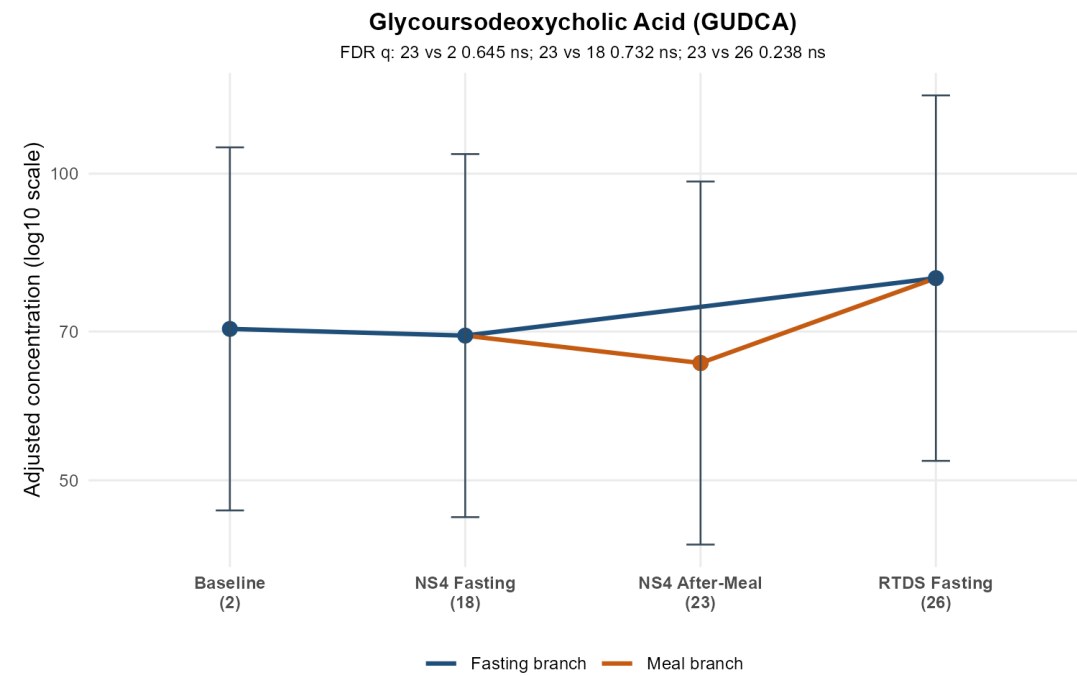
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	-0.184	0.83	0.411	0.561	ns
23 vs 18	-0.192	0.83	0.382	0.538	ns
23 vs 26	-0.090	0.91	0.691	0.819	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

GUDCA: Glycoursodeoxycholic Acid (GUDCA)

Tertiary / Microbial | Individual BA | N=126, subjects=32



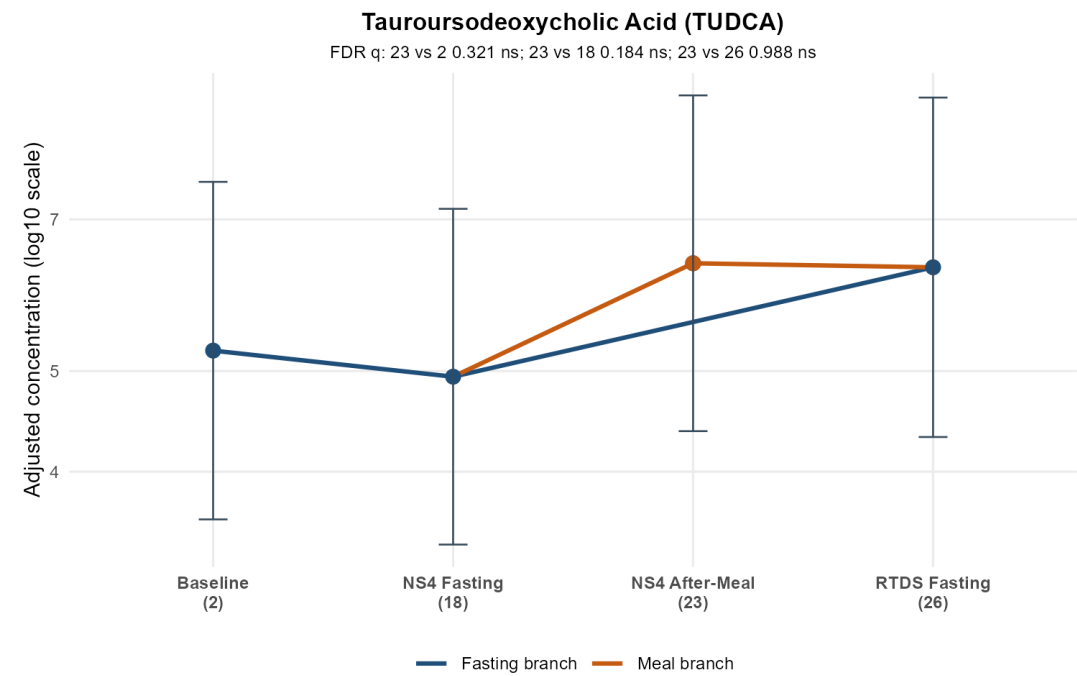
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	-0.077	0.93	0.516	0.645	ns
23 vs 18	-0.062	0.94	0.602	0.732	ns
23 vs 26	-0.192	0.83	0.116	0.238	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

TUDCA: Tauroursodeoxycholic Acid (TUDCA)

Tertiary / Microbial | Individual BA | N=125, subjects=32



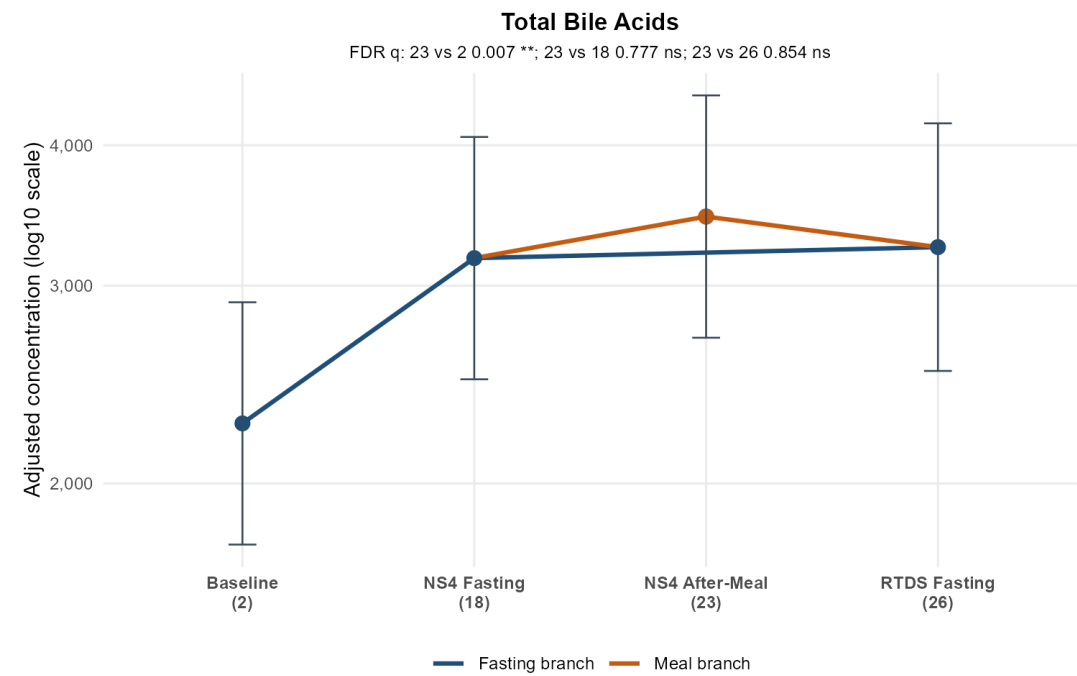
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.194	1.21	0.171	0.321	ns
23 vs 18	0.252	1.29	0.074	0.184	ns
23 vs 26	0.009	1.01	0.950	0.988	ns

Divergence: the meal branch starts from 18 and merges at 26; only point 23 is colored orange. Points are LMM EMMs; error bars are 95% CIs.

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

Total_BA: Total Bile Acids

Aggregate Total Pool | Aggregate class | N=126, subjects=32



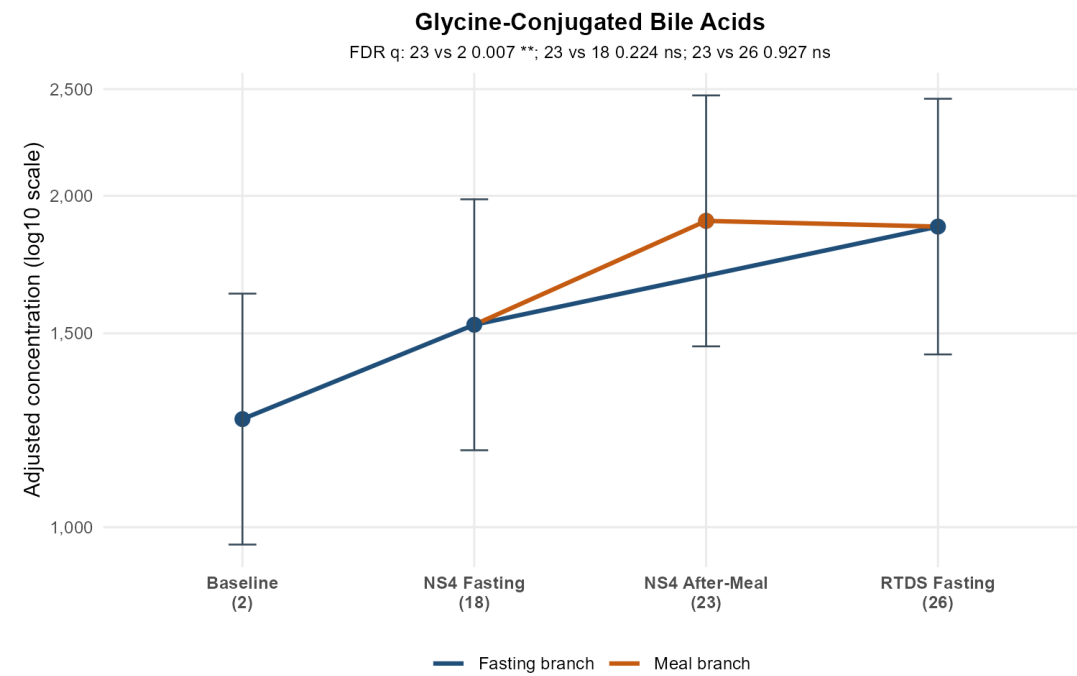
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.424	1.53	0.002	0.007	**
23 vs 18	0.085	1.09	0.518	0.777	ns
23 vs 26	0.063	1.06	0.641	0.854	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=0.007, **).

Aggregate-class inference is derived from summed component bile acids after preserving all-missing rows as missing. These aggregate tests should be interpreted as class-level summaries, not replacements for the individual bile-acid results.

Glycine_Conjugated: Glycine-Conjugated Bile Acids

Aggregate Glycine Conjugates | Aggregate class | N=126, subjects=32



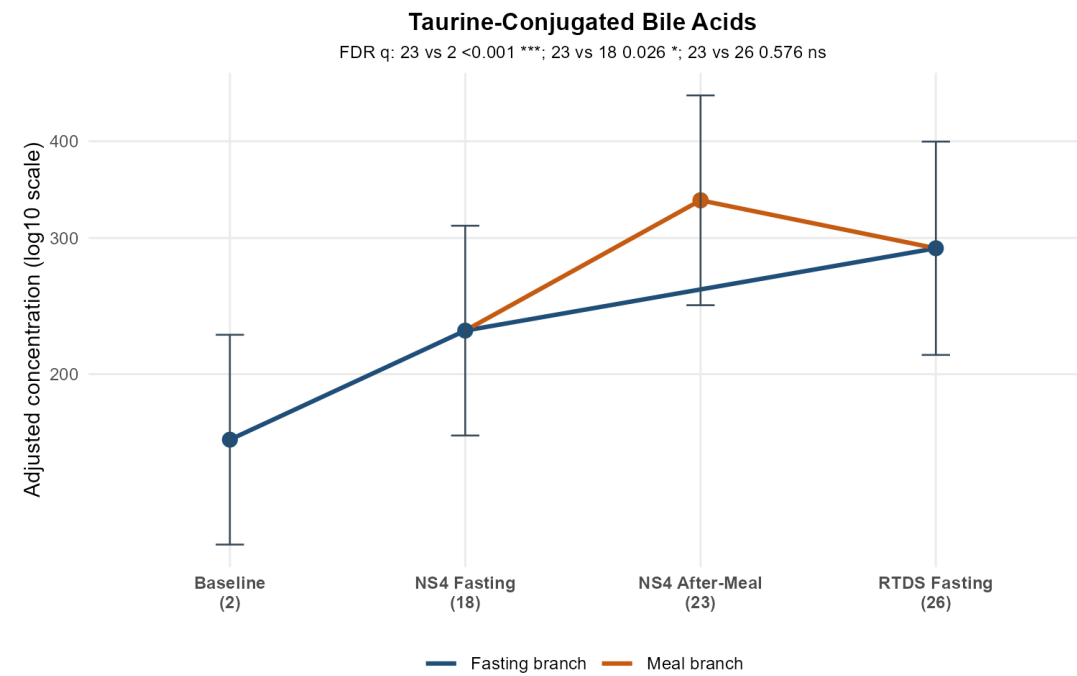
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.415	1.51	0.002	0.007	**
23 vs 18	0.217	1.24	0.093	0.224	ns
23 vs 26	0.012	1.01	0.927	0.927	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=0.007, **).

Aggregate-class inference is derived from summed component bile acids after preserving all-missing rows as missing. These aggregate tests should be interpreted as class-level summaries, not replacements for the individual bile-acid results.

Taurine_Conjugated: Taurine-Conjugated Bile Acids

Aggregate Taurine Conjugates | Aggregate class | N=126, subjects=32



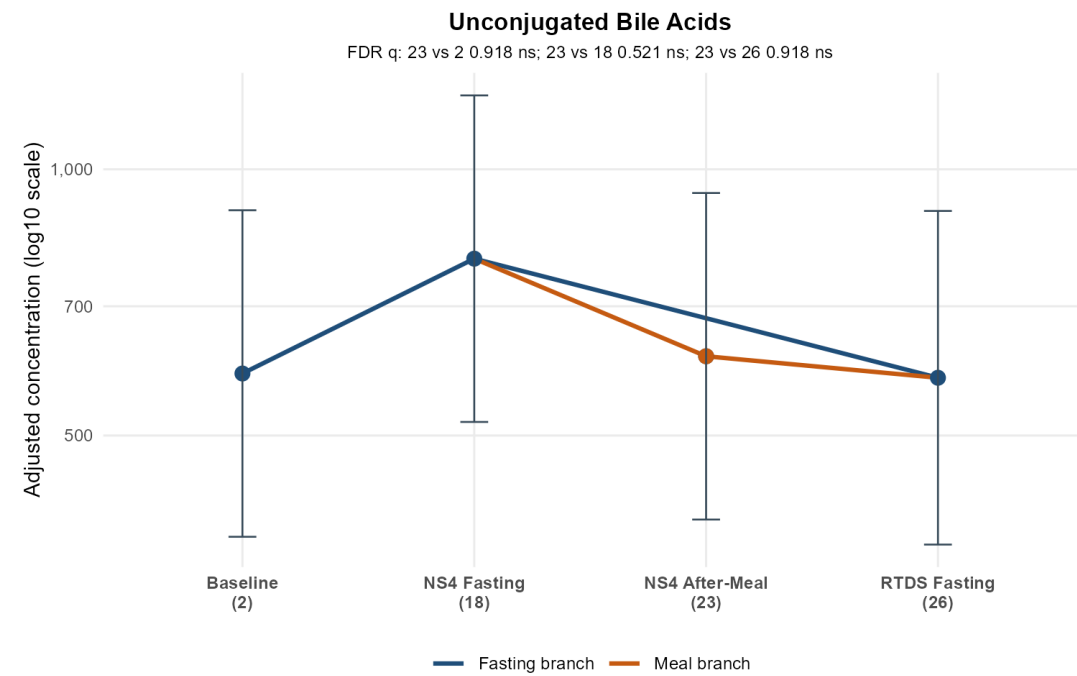
Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.713	2.04	<0.001	<0.001	***
23 vs 18	0.388	1.47	0.009	0.026	*
23 vs 26	0.143	1.15	0.336	0.576	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was higher than baseline fasting (2) (FDR q=<0.001, ***); higher than NS4 fasting (18) (FDR q=0.026, *).

Aggregate-class inference is derived from summed component bile acids after preserving all-missing rows as missing. These aggregate tests should be interpreted as class-level summaries, not replacements for the individual bile-acid results.

Unconjugated: Unconjugated Bile Acids

Aggregate Unconjugated Pool | Aggregate class | N=126, subjects=32



Contrast	LFC	Fold	raw p	FDR q	Sig
23 vs 2	0.045	1.05	0.842	0.918	ns
23 vs 18	-0.254	0.78	0.261	0.521	ns
23 vs 26	0.056	1.06	0.808	0.918	ns

FDR-based interpretation: After BH FDR correction, timepoint 23 was not significantly different from baseline fasting, NS4 fasting, or RTDS fasting for this outcome.

Aggregate-class inference is derived from summed component bile acids after preserving all-missing rows as missing. These aggregate tests should be interpreted as class-level summaries, not replacements for the individual bile-acid results.

Reproducibility Files

File	Purpose
comparison_work/fixed_csv_lmm_analysis.R	Recomputes corrected LMM results, q-values, verification checks, and plots.
comparison_work/fixed_csv_report_outputs/corrected_lmm_results.csv	Primary corrected result table used for this PDF.
comparison_work/fixed_csv_report_outputs/corrected_emm_results.csv	Back-transformed estimated marginal means and confidence intervals.
comparison_work/fixed_csv_report_outputs/verification_checks.csv	Audit checks printed in this report.
comparison_work/fixed_csv_report_outputs/session_info.txt	R version and package session details.